

Fourier analysis – 3

The Fourier transform on the line

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Theoretical minimum.

Definition. For $f \in L^1(\mathbb{R})$ put $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-ix\xi}dx$. Then $\widehat{f}$ is continuous, bounded by $\|f\|_1$ and tends to zero at infinity (Riemann–Lebesgue). Differentiation and multiplication are exchanged: $\widehat{f'}(\xi) = i\xi\widehat{f}(\xi)$ and $\widehat{xf}(\xi) = i\widehat{f'}(\xi)$, so smoothness of f is decay of $\widehat{f}$ and vice versa.

Theorem 1 (inversion and Plancherel). If f and $\widehat{f}$ are both integrable, then $f(x) = \frac{1}{2\pi} \int \widehat{f}(\xi)e^{ix\xi}d\xi$ almost everywhere. On the Schwartz class the transform is a bijection, and $\int |f|^2 = \frac{1}{2\pi} \int |\widehat{f}|^2$ (Plancherel), so it extends to a unitary map of $L^2(\mathbb{R})$ up to the factor $\sqrt{2\pi}$.

Theorem 2 (convolution and the Gaussian). $\widehat{f * g} = \widehat{f} \cdot \widehat{g}$; the Gaussian $e^{-x^2/2}$ is a fixed point up to the factor $\sqrt{2\pi}$, and the family $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$ is an approximate identity. Convolution with a smooth kernel smooths, multiplication by a cutoff localises, and the transform turns each of these operations into the other.

Theorem 3 (Poisson summation). For a Schwartz function $\sum_{n \in \mathbb{Z}} f(n) = \sum_{k \in \mathbb{Z}} \widehat{f}(2\pi k)$. The identity was proved in Complex analysis – 4; here it is used as a tool, for instance to derive the functional equation of the theta function $\theta(t) = \sum e^{-\pi n^2 t}$.

Fact 1 (the uncertainty principle). A function and its transform cannot both be concentrated: $(\int x^2 |f|^2)(\int \xi^2 |\widehat{f}|^2) \geq \frac{1}{16\pi^2} (\int |f|^2)^2$ in the normalisation of problem 4 below, with equality only for Gaussians; and a non-zero compactly supported function can never have a compactly supported transform.

1. Prove the basic rules: $\widehat{f'} = i\xi\widehat{f}$, $\widehat{xf} = i\widehat{f'}$, $\widehat{f*g} = \widehat{f}\widehat{g}$, the behaviour under translation and dilation; compute the transforms of $e^{-|x|}$, $\frac{1}{1+x^2}$, the indicator of $[-1, 1]$ and the triangle function, and check the inversion formula on them.
2. Prove that $e^{-x^2/2}$ is its own Fourier transform up to a constant, deduce Plancherel's identity on the Schwartz class, and use the transform to evaluate $\int_0^\infty \frac{\sin x}{x} dx$ and $\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx$.
3. Prove the Poisson summation formula for Schwartz functions and deduce the functional equation $\theta(1/t) = \sqrt{t}\theta(t)$ for $\theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}$.
4. (S.T. Yau CSMC 2014) Let $f \in L^2(\mathbb{R})$ and let $\widehat{f}$ be its Fourier transform. Prove that

$$\int_{-\infty}^{\infty} x^2 |f(x)|^2 dx \int_{-\infty}^{\infty} \xi^2 |\widehat{f}(\xi)|^2 d\xi \geq \frac{\left(\int_{-\infty}^{\infty} |f(x)|^2 dx\right)^2}{16\pi^2},$$

under the convergence conditions that make both sides finite, and determine the case of equality.

5. (S.T. Yau CSMC 2019, team) Show that there is no non-zero compactly supported smooth function $f \in C_0^\infty(\mathbb{R}^2)$ whose Fourier transform $\widehat{f}$ is also compactly supported.